(a) Expert-Driven Curation

1. Sources
Top Journals, Conferences & Popular databases

2. ML + Biology Experts
Data filtering, selection, preprocessing, and splitting

3. Unified Public Task Capsules
Train/test data, task description, sample submission.

4. Hidden Private Labels
Ground truth labels, grading function.

(b) 9 Biomedical Domains, 76 Tasks

- Sequence
- Single Cell
- 3D Structure
- Network Biology
- Chemical Biology
- Perturbation Dynamics
- Phenotype Disease
- Imaging Integrated
- Text Integrated

(c) Evaluating 18 Agent Configurations

7 Minimal Configurations
$Minimal_{g56s}$ $Minimal_{o46}$
$Minimal_{ge}$ $Minimal_{glm}$
$Minimal_{qw}$ $Minimal_{gm}$
$Minimal_{dv3}$

7 Generic Agent Configs
$OpenCode_{g56s}$ $OpenCode_{o46}$
$Codex_{g56s}$ $Codex_{g6a}$
$Claude\ Code_{o46}$ $Claude\ Code_{o5}$
$Claude\ Code_{o55}$

2 Biomedical Agents
$Biomni_{s5}$
$STELLA_{s5}$

2 ML Coding Agents
$MLEvolve_{ge}$
$MLMaster2_{dv4}$

(d) Execution

Sandbox Resources
Configurable time/GPU budget; unified conda environment

Output
Submission.csv → Held-out grader

(e) Metrics & Analysis

Evaluation Metrics (9 Types)
Pearson / Spearman / Accuracy
Macro-F1 / ROC-AUC / AUPRC
C-index / Dice / TM-score

Analysis Output
Leaderboard | Domain Heatmap
Failure Taxonomy | Cost Analysis